

Lecture 7

Lecturer: Siyue Liu

Scribe: ChatGPT

Convention. The ground set is E , and $M = (E, \mathcal{I})$ denotes a matroid with bases $\mathcal{B}(M)$. We write $S - e + u = (S \setminus \{e\}) \cup \{u\}$, $\chi^S \in \{0, 1\}^E$ for the incidence vector of S , and $S \triangle T$ for symmetric difference. Unless stated otherwise, $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$ is submodular. Monotonicity is assumed only where it is invoked.

1 Why Local Search?

Local search starts from a feasible set, examines a prescribed neighborhood, and moves whenever a neighbor has larger value. The neighborhood must be large enough to admit useful exchange structure and small enough to search. For the intersection of $k \geq 2$ matroids, the classical and local-search bounds are as follows.

Objective and constraints	Classical guarantee	Local-search guarantee
Monotone submodular, k matroids	$1/(k+1)$	$1/(k+\varepsilon)$
Linear, k matroids	$1/k$	$1/(k-1+\varepsilon)$

The classical bounds follow from greedy analysis; the local-search bounds, for $k \geq 2$, are due to Lee, Sviridenko, and Vondrák [3, 4]. We first develop the one-matroid argument directly, then isolate the fractional certificate underlying the more general analysis.

2 One-Swap Search for One Matroid

Assume in this section that f is monotone. Any independent set can be extended to a basis without decreasing its value, so an optimal solution may be taken to be a basis. Let r be the rank of M and fix an optimal basis OPT .

Definition 2.1 (One-swap local optimum). A basis $S \in \mathcal{B}(M)$ is an *exact one-swap local optimum* if

$$f(S - e + u) \leq f(S)$$

for every $e \in S$ and $u \in E \setminus S$ for which $S - e + u \in \mathcal{B}(M)$. For $\eta > 0$, it is an η -*multiplicative one-swap local optimum* if every such exchange satisfies

$$f(S - e + u) \leq (1 + \eta)f(S).$$

Algorithm 1 Threshold one-swap local search

Input: A matroid $M = (E, \mathcal{I})$ of positive rank, a value oracle for f , and $\eta \geq 0$.

Choose a nonloop u^* maximizing $f(\{u\})$, and extend $\{u^*\}$ to a basis S .

while there are $e \in S$ and $u \in E \setminus S$ such that

$S - e + u \in \mathcal{B}(M)$ and $f(S - e + u) > (1 + \eta)f(S)$ **do**

$S \leftarrow S - e + u$.

end while

return S .

The choice $\eta = 0$ gives exact local search, which terminates because there are finitely many bases, but it need not take polynomially many iterations. For $\eta > 0$, every successful move increases the value multiplicatively.

Lemma 2.2 (Iteration bound). *If $\eta > 0$ and $f(\text{OPT}) > 0$, Algorithm 1 makes at most*

$$\left\lceil \log_{1+\eta} r \right\rceil$$

successful exchanges. A direct neighborhood scan uses at most $r(|E| - r)$ independence and value-oracle tests.

Proof. Submodularity, nonnegativity, and monotonicity give

$$f(\text{OPT}) \leq \sum_{e \in \text{OPT}} f(\{e\}) \leq rf(\{u^*\}) \leq rf(S_0),$$

where S_0 is the initial basis. Every accepted exchange multiplies the current value by more than $1 + \eta$, while every iterate is a basis and hence has value at most $f(\text{OPT})$. Thus $(1 + \eta)^t f(S_0) < f(\text{OPT}) \leq rf(S_0)$ after t moves. If $f(S_0) = 0$, the displayed bound implies $f(\text{OPT}) = 0$, and the optimization problem is already trivial. $\square$

The structural ingredient in the approximation analysis is a one-sided bijective exchange theorem.

Lemma 2.3 (Bijective basis exchange [2]). *For bases $B, B' \in \mathcal{B}(M)$, there is a bijection $\pi : B \rightarrow B'$ such that*

$$\pi(e) = e \quad (e \in B \cap B'), \quad B - e + \pi(e) \in \mathcal{B}(M) \quad (e \in B).$$

Remark. The lemma is one-sided: it guarantees that the exchanged version of B is a basis. It does not assert that one bijection simultaneously makes $B' - \pi(e) + e$ a basis for every e .

Theorem 2.4 (Exact one-swap analysis). *Every exact one-swap local optimum S satisfies*

$$2f(S) \geq f(\text{OPT}) + f(\emptyset).$$

In particular, $f(S) \geq \frac{1}{2}f(\text{OPT})$.

Proof. Let $\pi : S \rightarrow \text{OPT}$ be the bijection above. Local optimality, or equality when $\pi(e) = e$, implies

$$f(\pi(e) \mid S - e) \leq f(e \mid S - e) \quad (e \in S).$$

Write $S = \{e_1, \dots, e_r\}$ and $S_i = \{e_1, \dots, e_i\}$. Monotonicity and diminishing returns give

$$\begin{aligned}
f(\text{OPT}) - f(S) &\leq f(S \cup \text{OPT}) - f(S) \\
&\leq \sum_{e \in S} f(\pi(e) \mid S) \\
&\leq \sum_{e \in S} f(\pi(e) \mid S - e) \\
&\leq \sum_{e \in S} f(e \mid S - e) \\
&\leq \sum_{i=1}^r f(e_i \mid S_{i-1}) = f(S) - f(\emptyset).
\end{aligned}$$

Rearranging proves the claim. $\square$

The threshold in the handwritten algorithm must be scaled with the rank. A fixed multiplicative error can accumulate once for each paired exchange.

Theorem 2.5 (Approximate one-swap analysis). *The output of Algorithm 1 satisfies*

$$f(S) \geq \frac{f(\text{OPT}) + f(\emptyset)}{2 + r\eta} \geq \frac{f(\text{OPT})}{2 + r\eta}.$$

Consequently, for $0 < \varepsilon \leq 1$, taking $\eta = \varepsilon/r$ gives a $1/(2 + \varepsilon)$ -approximation in $O((r/\varepsilon) \log(r + 1))$ successful iterations.

Proof. At termination,

$$\begin{aligned}
f(\pi(e) \mid S - e) &\leq (1 + \eta)f(S) - f(S - e) \\
&= f(e \mid S - e) + \eta f(S).
\end{aligned}$$

Repeating the chain in Theorem 2.4 and summing the additional error over the r elements gives

$$f(\text{OPT}) - f(S) \leq f(S) - f(\emptyset) + r\eta f(S).$$

The result follows by rearranging and applying Lemma 2.2. $\square$

Remark (The threshold in the handwritten notes). Writing the acceptance threshold as $f(S') > f(S)/(1 - \delta)$ is equivalent to $1 + \eta = 1/(1 - \delta)$. To obtain the denominator $2 + \varepsilon$, take

$$\delta = \frac{\varepsilon}{r + \varepsilon}, \quad \eta = \frac{\delta}{1 - \delta} = \frac{\varepsilon}{r}.$$

3 A Fractional Local-Optimality Certificate

For a nonmonotone objective, extending a set to a basis can decrease its value. The natural one-exchange neighborhood for an arbitrary independent set therefore permits additions, deletions, and swaps.

Algorithm 2 Add–delete–swap local search

Input: A matroid $M = (E, \mathcal{I})$, a value oracle for f , and $S_0 \in \mathcal{I}$.

$S \leftarrow S_0$.

while there is $A \in \mathcal{I}$ with $|A \setminus S| \leq 1$, $|S \setminus A| \leq 1$, and $f(A) > f(S)$ **do**

$S \leftarrow A$.

end while

return S .

As written, Algorithm 2 describes an exact local optimum and may take exponentially many iterations. A multiplicative threshold gives a polynomial termination bound; the approximation loss must again account for how many local inequalities are combined.

We next prove a general device for combining such inequalities. Recall that the Lovász extension $\hat{f} : [0, 1]^E \rightarrow \mathbb{R}$ may be written as

$$\hat{f}(x) = \int_0^1 f(\{e \in E : x_e \geq \theta\}) d\theta.$$

It agrees with f at incidence vectors and is convex precisely when f is submodular [5].

Lemma 3.1 (Lattice submodularity of the Lovász extension). *For $x, y \in [0, 1]^E$,*

$$\hat{f}(x) + \hat{f}(y) \geq \hat{f}(x \wedge y) + \hat{f}(x \vee y),$$

where minimum and maximum are taken coordinatewise.

Proof. For each threshold θ , let $X_\theta = \{e : x_e \geq \theta\}$ and $Y_\theta = \{e : y_e \geq \theta\}$. The threshold sets of $x \wedge y$ and $x \vee y$ are respectively $X_\theta \cap Y_\theta$ and $X_\theta \cup Y_\theta$. Apply submodularity at every threshold and integrate from 0 to 1. $\square$

Theorem 3.2 (Fractional local-optimality certificate). *Let $\alpha \geq 1$, let $S, T, A_1, \dots, A_m \subseteq E$, and let $\lambda_1, \dots, \lambda_m \geq 0$. If*

$$\chi^T + (\alpha - 1)\chi^{S \cap T} = \alpha\chi^S + \sum_{i=1}^m \lambda_i(\chi^{A_i} - \chi^S), \quad (1)$$

then

$$(\alpha + 1)f(S) + \sum_{i=1}^m \lambda_i(f(A_i) - f(S)) \geq f(S \cup T) + \alpha f(S \cap T). \quad (2)$$

Proof. The assertion is invariant under replacing f by $f - c$, because both sides of (2) decrease by $(\alpha + 1)c$. We may therefore assume $f(\emptyset) = 0$.

Set $\lambda = \sum_i \lambda_i$. If $\lambda = 0$, (1) forces $S = T$, and the conclusion is an equality. Assume $\lambda > 0$. Rearranging the identity gives the convex combination

$$x := \sum_{i=1}^m \frac{\lambda_i}{\lambda} \chi^{A_i} = \frac{1}{\lambda} \chi^T + \frac{\alpha - 1}{\lambda} \chi^{S \cap T} + \left(1 - \frac{\alpha}{\lambda}\right) \chi^S. \quad (3)$$

Convexity of $\hat{f}$ yields

$$\sum_{i=1}^m \lambda_i f(A_i) \geq \lambda \hat{f}(x). \quad (4)$$

Write $I = S \cap T$ and $U = S \cup T$. The coordinates of x are

$$x_e = \begin{cases} 1, & e \in I, \\ 1 - \alpha/\lambda, & e \in S \setminus T, \\ 1/\lambda, & e \in T \setminus S, \\ 0, & e \notin U. \end{cases}$$

There are three possible orders of these levels.

If $\lambda \geq \alpha + 1$, the threshold formula gives

$$\lambda \widehat{f}(x) = f(U) + (\lambda - \alpha - 1)f(S) + \alpha f(I). \quad (5)$$

If $\alpha \leq \lambda \leq \alpha + 1$, it gives

$$\begin{aligned} \lambda \widehat{f}(x) &= (\lambda - \alpha)f(U) + (\alpha + 1 - \lambda)f(T) + (\lambda - 1)f(I) \\ &\geq f(U) + (\lambda - \alpha - 1)f(S) + \alpha f(I). \end{aligned}$$

Indeed, the difference between the two lines is

$$(\alpha + 1 - \lambda)(f(S) + f(T) - f(U) - f(I)) \geq 0.$$

Finally, suppose $\lambda < \alpha$. Since the left side of (3) belongs to $[0, 1]^E$, its allegedly negative coordinate $1 - \alpha/\lambda$ cannot occur; hence $S \setminus T = \emptyset$. Thus $S = I \subseteq T = U$. If $T \setminus S \neq \emptyset$, the coordinate $1/\lambda \leq 1$ also forces $\lambda \geq 1$, and the threshold formula gives

$$\lambda \widehat{f}(x) = f(T) + (\lambda - 1)f(S) = f(U) + (\lambda - \alpha - 1)f(S) + \alpha f(I).$$

If $T = S$, then $x = \chi^S$ and the same identity holds directly. Together with (4), all three cases imply

$$\sum_i \lambda_i f(A_i) \geq f(U) + \alpha f(I) + (\lambda - \alpha - 1)f(S).$$

Adding $(\alpha + 1 - \lambda)f(S)$ proves (2). □

3.1 The base-polytope interpretation

The matroid base polytope is

$$B(M) = \text{conv}\{\chi^B : B \in \mathcal{B}(M)\}.$$

Two vertices χ^S and χ^A are adjacent exactly when A is obtained from S by one exchange. Consequently, the tangent cone at χ^S is generated by the directions to one-exchange bases [6]. For every other basis T , there are one-exchange bases A_i and coefficients $\lambda_i \geq 0$ such that

$$\chi^T = \chi^S + \sum_i \lambda_i (\chi^{A_i} - \chi^S).$$

If S is an exact local optimum, then $f(A_i) \leq f(S)$. Applying Theorem 3.2 with $\alpha = 1$ gives

$$2f(S) \geq f(S \cup T) + f(S \cap T).$$

For monotone nonnegative f , the right side is at least $f(T)$. This recovers Theorem 2.4 from the geometry of $B(M)$.

4 Bounded Exchanges for Several Matroids

Let $M_j = (E, \mathcal{I}_j)$ for $j = 1, \dots, k$, and let $\mathcal{I} = \bigcap_{j=1}^k \mathcal{I}_j$. For an integer $p \geq 1$, a p -exchange may add at most $2p$ elements and delete at most $2kp$ elements.

Algorithm 3 p -exchange local search

Input: Matroids $M_1, \dots, M_k$, a value oracle for f , an integer $p \geq 1$, and $S_0 \in \mathcal{I}$.

$S \leftarrow S_0$.

while there is $A \in \mathcal{I}$ such that

$|A \setminus S| \leq 2p$, $|S \setminus A| \leq 2kp$, and $f(A) > f(S)$ **do**

$S \leftarrow A$.

end while

return S .

For fixed k and p , one neighborhood can be enumerated in polynomial time. Exact local search can still require exponentially many improving moves; a sufficiently fine multiplicative threshold makes the overall algorithm polynomial.

Theorem 4.1 (Generalized exchange local search [4]). *Let $k \geq 2$. If $S \in \mathcal{I}$ is an exact local optimum under p -exchanges, then every $T \in \mathcal{I}$ satisfies*

$$\left(k + \frac{1}{p}\right) f(S) \geq f(S \cup T) + \left(k - 1 + \frac{1}{p}\right) f(S \cap T).$$

After using multiplicative approximate local search and choosing $p = \Theta(1/\varepsilon)$, this yields polynomial-time guarantees

$$\frac{1}{k + \varepsilon} \quad \text{for monotone submodular objectives,} \quad \frac{1}{k - 1 + \varepsilon} \quad \text{for linear objectives.}$$

Remark (Why one more matroid becomes a submodular objective). Given $M_1, \dots, M_{k+1}$, maximize the rank function $r_{M_{k+1}}(S)$ subject to $S \in \bigcap_{j=1}^k \mathcal{I}_j$. The rank function is monotone submodular. From any feasible S , retain a largest subset independent in M_{k+1} ; its size is $r_{M_{k+1}}(S)$ and it remains independent in the first k matroids. Thus a ρ -approximation for monotone submodular maximization under k matroids also gives a ρ -approximation for cardinality intersection of $k + 1$ matroids.

5 Matching as a Model Case

Let S and T be matchings in a graph. Every component of $S \Delta T$ is an alternating path or an alternating even cycle. This simple decomposition is the prototype for the generalized exchange properties used in Theorem 4.1. In a bipartite graph, matchings are exactly the common independent sets of two partition matroids.

5.1 Cardinality: short augmenting paths

Lemma 5.1 (A short augmenting path). *Let $\varepsilon > 0$. If $|T| > (1 + \varepsilon)|S|$, then $S \Delta T$ contains an S -augmenting path with fewer than*

$$1 + \frac{2}{\varepsilon}$$

edges.

Proof. Let a and b be the numbers of path components having respectively one more T -edge and one more S -edge. Then $|T| - |S| = a - b$, so $a \geq |T| - |S|$. The a S -augmenting paths together contain at most $|S| + |T|$ edges. One of them therefore has length at most

$$\frac{|S| + |T|}{a} \leq \frac{|S| + |T|}{|T| - |S|} = 1 + \frac{2|S|}{|T| - |S|} < 1 + \frac{2}{\varepsilon}.$$

□

This is a quantitative form of the augmenting-path principle [1]. If no augmenting path of length at most $\lceil 1 + 2/\varepsilon \rceil$ exists, then S is a $(1 + \varepsilon)$ -approximation to a maximum-cardinality matching.

5.2 A fractional family of bounded exchanges

For a submodular objective, a single augmenting path need not capture enough value. Instead, we average many bounded alternating exchanges.

Lemma 5.2 (Bounded matching exchanges). *Fix an integer $h \geq 1$. For any matchings S, T , there are matchings $A_1, \dots, A_m$ and weights $\lambda_i \geq 0$ such that*

$$|A_i \triangle S| \leq 2h + 1 \quad (i = 1, \dots, m)$$

and

$$\chi^T + \frac{1}{h} \chi^{S \cap T} = \left(1 + \frac{1}{h}\right) \chi^S + \sum_{i=1}^m \lambda_i (\chi^{A_i} - \chi^S). \quad (6)$$

Moreover, the exchanges can be chosen so that $\sum_i \lambda_i \leq 3|E|$.

Proof. Decompose $S \triangle T$ into alternating paths and cycles. On a path, list its T -edges in order. For each of the h possible offsets, cut this list every h positions. Every resulting block contains at most h T -edges. Exchange a block together with all incident S -edges in that alternating component, and give the exchange weight $1/h$. It changes at most h edges of $T \setminus S$ and at most $h + 1$ edges of $S \setminus T$, so its radius is at most $2h + 1$.

Across the h offsets, each T -edge is used h times. An internal S -edge is used once in every offset, and once more in the unique offset whose cut separates its two neighboring T -edges. Its total weighted multiplicity is therefore $1 + 1/h$. An endpoint S -edge receives the missing weight $1/h$ from a singleton deletion. A component consisting of one isolated S -edge is deleted with weight $1 + 1/h$.

For an alternating cycle longer than $2h$, take every cyclic block of h consecutive T -edges with weight $1/h$. If the cycle has at most $2h$ edges, use the whole-cycle exchange with weight 1 and give each of its S -edges an additional singleton-deletion weight $1/h$. Thus every edge of $T \setminus S$ has total positive multiplicity 1, while every edge of $S \setminus T$ has total negative multiplicity $1 + 1/h$. Summing the exchange vectors gives

$$\sum_i \lambda_i (\chi^{A_i} - \chi^S) = \chi^{T \setminus S} - \left(1 + \frac{1}{h}\right) \chi^{S \setminus T},$$

which is equivalent to (6). The construction uses total weight at most $3|E|$. □

The factor $(1 + 1/h)$ multiplying χ^S in (6) is essential. Omitting it, as in the first line of the handwritten sketch, makes the incidence-vector identity false.

Algorithm 4 Bounded-exchange local search for a matching

Input: A graph, a value oracle for monotone submodular f , an integer $h \geq 1$, and a threshold $\tau \geq 0$.
Choose a matching S containing a feasible edge of maximum singleton value.

while there is a matching A with $|A \Delta S| \leq 2h + 1$ and

$f(A) > (1 + \tau)f(S)$ **do**

$S \leftarrow A$.

end while

return S .

Theorem 5.3 (Submodular matching). *If $\tau = 0$, the output S of Algorithm 4 satisfies, for every matching T ,*

$$\left(2 + \frac{1}{h}\right) f(S) \geq f(S \cup T) + \left(1 + \frac{1}{h}\right) f(S \cap T) \geq f(T).$$

Hence $h \geq 1/\varepsilon$ gives a $1/(2 + \varepsilon)$ -approximation. For every fixed $\varepsilon > 0$, a suitably thresholded implementation runs in polynomial time [4].

Proof. Apply Theorem 3.2 to (6) with $\alpha = 1 + 1/h$. Exact local optimality makes every term $f(A_i) - f(S)$ nonpositive, which proves the first inequality. Monotonicity and nonnegativity give $f(S \cup T) + (1 + 1/h)f(S \cap T) \geq f(T)$.

For the polynomial version, fix $0 < \varepsilon \leq 1$ and choose $h = \lceil 2/\varepsilon \rceil$ and $\tau = \varepsilon/(6|E|)$. Lemma 5.2 bounds the accumulated local-optimality error by $3|E|\tau f(S) \leq (\varepsilon/2)f(S)$, while $1/h \leq \varepsilon/2$. The neighborhood has size $|E|^{O(1/\varepsilon)}$, and multiplicative improvements give polynomially many iterations for fixed ε . Indeed, submodularity and the maximum-singleton initialization give $f(T^*) \leq |E|f(S_0)$ for an optimal matching T^* . Thus, unless the optimum is zero, at most $\lceil \log_{1+\tau} |E| \rceil = O((|E|/\varepsilon) \log |E|)$ moves are accepted. $\square$

Summary. For one matroid, a bijective basis exchange turns one-swap optimality into a $1/2$ guarantee. The Lovász-extension certificate packages a weighted family of local inequalities into one global inequality. Generalized matroid exchanges then yield the $1/(k + \varepsilon)$ local-search bound, and alternating paths and cycles make the same mechanism concrete for matchings.

References

- [1] C. Berge, “Two theorems in graph theory,” *Proceedings of the National Academy of Sciences of the United States of America* 43 (1957), 842–844. doi:10.1073/pnas.43.9.842.
- [2] R. A. Brualdi, “Comments on bases in dependence structures,” *Bulletin of the Australian Mathematical Society* 1 (1969), 161–167. doi:10.1017/S000497270004140X.
- [3] M. L. Fisher, G. L. Nemhauser, and L. A. Wolsey, “An analysis of approximations for maximizing submodular set functions—II,” in *Polyhedral Combinatorics*, Mathematical Programming Studies 8, Springer, 1978, pp. 73–87. doi:10.1007/BFb0121195.
- [4] J. Lee, M. Sviridenko, and J. Vondrák, “Submodular maximization over multiple matroids via generalized exchange properties,” *Mathematics of Operations Research* 35 (2010), 795–806. doi:10.1287/moor.1100.0463.
- [5] L. Lovász, “Submodular functions and convexity,” in *Mathematical Programming: The State of the Art—Bonn 1982*, A. Bachem, B. Korte, and M. Grötschel, eds., Springer, 1983, pp. 235–257. doi:10.1007/978-3-642-68874-4_10.
- [6] A. Schrijver, *Combinatorial Optimization: Polyhedra and Efficiency*, Algorithms and Combinatorics 24, Springer, 2003. <https://link.springer.com/book/9783540443896>.